MODULE 5

1.

What are the different techniques for interfacing I/O device with CPU?

Memory mapped I/O and I/O mapped I/O

2.

. What is memory mapped I/O?

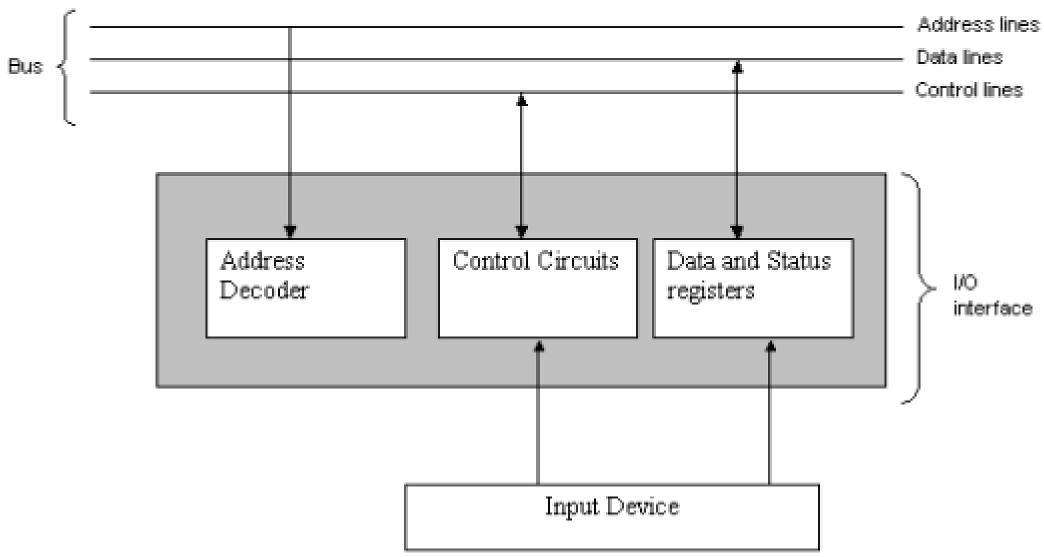
This is one of the techniques for interfacing I/O devices with CPU. In memory mapped I/O, the I/O devices assigned and identified by 16-bit addresses. To transfer the data between CPU and I/O devices memory related instructions (such as LDA, STA etc.) and memory control signals (MEMR, MEMW) are used.

The CPU selects an I/O device by placing the address on the address lines; the corresponding I/O device recognizes its address and responds to the CPU's command. Any machine instruction used to access memory can be used to transfer data to and from I/O devices.

Example:

MOVE DATAIN, R_0

MOVE R_0, DATAOUT, Where DATAIN and DATAOUT are I/O buffer registers.



3.

What is I/O mapped I/O?

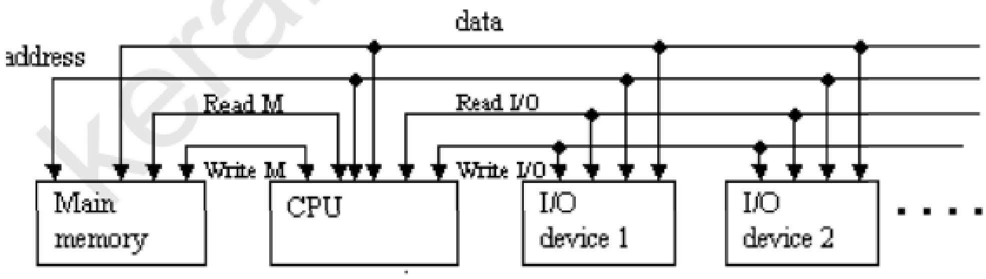
The memory and I/O address space are kept separate by Read/Write control lines.A memory reference instruction activates the Read M or Write M line and does not affect the I/O devices.

The I/O instruction such as for example IN or OUT activates the Read I/O and write I/O line and causes a word to be transferred between the addressed I/O and the CPU.

Therefore, the memory location and the I/O device may have the same address.

OUT data, device

IN data, device



4.

What are the different data transfer modes between the CPU and the I/O devices.

- Programmed I/O
- Interrupt driven I/O
- Direct Memory Access (DMA)

5.

List the advantages and disadvantages of Programmed I/O.

Advantages

- The programmed I/O mode is very low cost interface
- Can meet speeds of most of the peripherals.

Disadvantages: CPU speed is brought down to that of the slow peripheral device

6.

Explain Programmed I/O.

It is a mode of data transfer between the CPU and the logic within the peripherals. The transfer is initiated using an I/O transfer (IOT) instruction and controlled entirely by the CPU.

Programmed I/O can be:

– Synchronous (Or Unconditional) transfer: Here no check is made by the CPU to determine the state of the peripherals like Ready or Busy. The transfer takes place when ever the CPU directs it i.e. the I/O device must be ready to receive the output data from the CPU or to input data into the CPU when ever CPU initiates such a transfer. Hence the transfer is Unconditional. The data transfers are controlled by a clock of fixed frequency, independent of the CPU

– Asynchronous (Or Conditional) transfer: The peripheral is queried about its status to see if it is ready to perform the transfer. The actual transfer takes place only when the peripheral shows Ready state. This method accomplishes synchronization between the CPU and slower peripherals whose timing may not be fixed.

7.

Explain Interrupt driven I/O.

Here, the CPU does not wait for the peripheral device to become ready for the data transfer instead, it initiates the device and then goes away to handle other tasks. The peripheral sends the Ready signal to the CPU whenever it is ready. An interrupt request from an I/O device may cause the processor to suspend the execution of one program and start the execution of another program. Interrupts may sometimes alter the sequence of events. The programmer should be able to enable and disable interrupts. This is performed by machine instructions such as Interrupt Enable and Interrupt Disable.

8.

Explain the general sequence of events in interrupt I/O

Interrupt request - The I/O requesting peripheral sends an interrupt request signal (INT REQ) to the CPU to indicate that it is ready for an I/O operation

Interrupt acknowledge - On receiving the interrupt request signal, the CPU completes the execution of the current instruction and sends an Interrupt acknowledgement signal (INT ACK) to the interrupt requesting peripheral

Status saving - Before servicing the interrupt request, the return address of the main program is stored in some pre-assigned memory locations, so that the CPU can resume its normal operation on completion of the subroutine.

Device identification - The interrupt request to the CPU is only through a single line bus, if the system has more than one device connected to it, the CPU must identify the interrupting peripheral, so that it can call the appropriate interrupt service routine (ISR) from the memory.

Peripheral service – having identified the peripheral which requested for the service, the program branches to the starting address of the specific subroutine in the memory and it services the peripheral.

Restore CPU status – after completion of the service subroutine, the data from the CPU registers that were held temporarily in memory are reloaded back to their respective locations and the CPU can take up its normal task

Resume main task – when all registers and program counter are restored with their original contents, the main program operation is resumed.

9.

. Explain interrupt nesting.

An interrupt request from a high priority device should be accepted while the processor is servicing another request from a low priority device. For this the priority level of the processor can be set to that of the device that is currently being serviced. Interrupt requests from the devices with priority greater than that of the processor will only be accepted.

Processor's priority is encoded in a few bits of the Program Status Word. Hardware is implemented by using a separate INTR and INTA line for each device. An interrupt request is sent to the priority arbitration logic and is accepted only if it is greater than the one that is currently assigned to the processor.

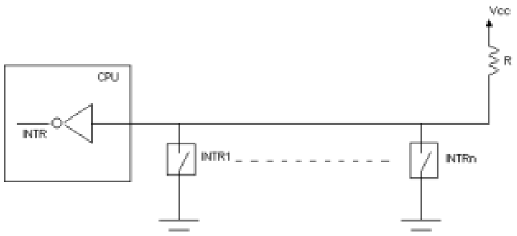
10.

Explain the methods for identifying the interrupting device.

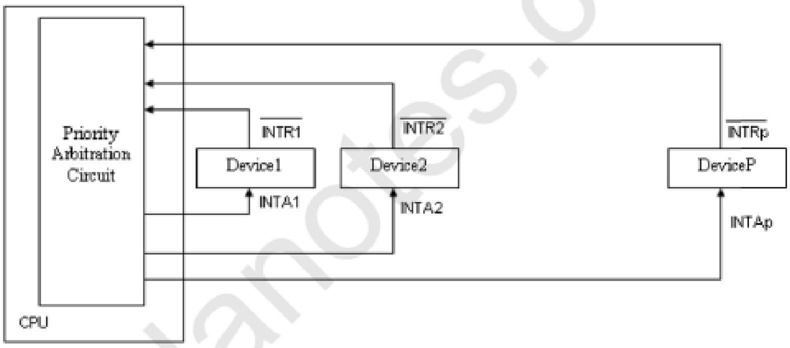
Polling and vectored interrupts are the methods to identifying the interrupting device.

Polling

All the devices are connected to a common INTR (Interrupt Request) line via switches to ground. When all the interrupt requests signals (INTR1 to INTRn) are inactive, the INTR line is maintained in the high voltage state. When any device requests an interrupt it closes its switch and the voltage on the line drops to zero, causing INTR line to go to 1.



A device that requests an interrupt identifies itself to the processor by sending a special code called vector to the processor. This code supplies the starting address of the ISR for that device. The processor reads this address, called the interrupt vector & loads into PC. This simplifies the circuit but it limits the number of devices that can be identified. One solution is to assign a code to a group of devices and the ISR identify the device by polling the devices in that group.



11.

What is DMA?

Direct memory access refers to that data transfer, when large blocks of data are to be transferred at high speeds between I/O devices and main memory, without involving the CPU. In order to perform DMA transfer, an additional interface circuitry is required known as **DMA controller.**

12.

Describe DMA Controller registers.

The DMA controller has the following four registers to perform transfer:

a. **Address register** – used to hold the address of the next word to be transferred
b. **Data buffer register** – used to store the data to be transferred
c. **Data count register or word counter** – used to store the number of words that are remaining to be transferred. It is automatically decremented after the transfer of each word and tested for zero. When the data count reaches zero, the DMA transfer halts.
d. **Control register and circuits** – used to control the DMA transfer

13.

. What is the need for DMA transfer?

During the actual transfer between peripheral and memory, CPU is free from this activity and can do some other useful work.

14.

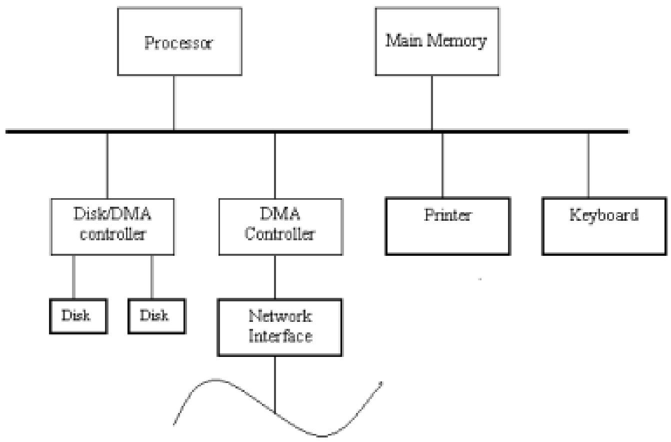
Explain the action carried out by the processor after occurrence of an interrupt.

An interrupt is a signal that refers to the transfer of program control from a currently running program to another service program as a result of an external or internal generated request. The control returns to the original program after the service program is executed. Interrupts are used to coordinate slow external devices with fast CPU. One of the bus control lines called the interrupt request line is dedicated for this purpose. Here, the CPU does not wait for the peripheral device to become ready for the data transfer instead, it initiates the device and then goes away to handle other tasks. The peripheral sends the Ready signal to the CPU whenever it is ready.

15.

Explain with the block diagram the DMA transfer in a computer system

The peripheral devices initiate the DMA transfer by sending a DMA request signal to the CPU when it is ready to perform a transfer.



There are two methods by which DMA transfer is initiated and operated:

Burst Mode

When the data is being transferred to the memory or from the memory, the CPU remains idle for relatively long periods of time or the CPU is in a kind of HOLD state. The CPU exits form this state only after the DMA request signal is withdrawn from the I/O device.The hold duration depends on the speed of the memory, I/O device, and number of bytes to be transferred.

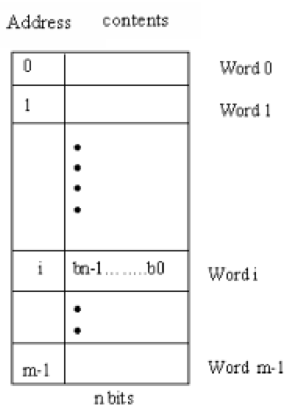
Cycle-Stealing Mode

The CPU after receiving the DMA request signal from the peripheral, will attend to the current instruction, it is doing and suspends its operation for the next time slice (machine cycle). During the next time slice, the DMA transfer takes place and one word is transferred between the peripheral device and the main memory. On completion of the transfer, the CPU can resume to its normal operation. DMA transfers are essentially time-sliced with the operations of the main memory. Here, the DMA controller can be regarded as "stealing" memory cycles from the CPU. Hence, this technique is known as cycle stealing.

16.

Explain memory system.

The main memory consists of a large number of storage cells, each of which can store a binary digit 0 or 1. This 1-bit representation of information is too small to be handled by a computer. So, a group of n bits are used while storing or retrieving. Each group of n bits is referred to as its word length. The word length can be 8 bits, 16 bits, 32, 64 or 128 bits depending on the size of the computer.



Each location in the main memory is given a distinct name or address. With this address it is possible to store or retrieve information from the memory location. The contents of memory locations can represent either instructions or operands.

17.

Explain Big Endian and Little Endian Assignments.

Two ways that byte address can be assigned across words. Big Endian and Little Endian. Big-endian is an order in which the "big end" (most significant value in the sequence) is stored first (at the lowest storage address). In big endian, store the most significant byte in the smallest address.

Word address	Byte address			
0	0	1	2	3
4	4	5	6	7
8	8	9	10	11

Little-endian is an order in which the "little end" (least significant value in the sequence) is stored first. In little endian, you store the least significant byte in the smallest address.

Word address	Byte address			
0	3	2	1	0
4	7	6	5	4
8				

18.

, Define Memory access time, memory cycle time and transfer rate.

- **Memory Access time:** It is the time that elapses between the initiation of an operation and the completion of that operation. E.g.: time between Read and MFC
- **Memory Cycle time :** It is the minimum time delay required between the initiations of two independent memory operations. E.g.: two successive Read operations.

- **Transfer Rate:** It is the rate at which the data can be transferred to or out of memory unit.

TN=TA+N/R

TN= Average time to read / write n bits

TA=- Average access time

N= Number of bits

R= Transfer rate in bits/ sec.

19.

). Discuss the Classification of memory

Memory can be either primary memory or Secondary memory.

Two classifications of primary storage are read-only memory (ROM) and random-access memory (RAM).

a) READ-ONLY MEMORY (ROM)

In computers, it is useful to have instructions that are used often, permanently stored inside the computer. ROM enables us to do this without loosing the programs and data when the computer is powered down. Only the computer manufacturer can provide these programs in ROM; once done, we cannot change it. Consequently, we cannot put any of our own data or programs in ROM. Many complex functions, such as translators for high-level languages, and operating systems are placed in ROM memory. Since these instructions are hardwired, they can be performed quickly and accurately.

- **Programmable ROM (PROM):** This is a type of ROM that can be programmed using special equipment; it can be written to, but only once. This is useful for companies that make their own ROMs from software they write, because when they change their code they can create new PROMs without requiring expensive equipment. This is similar to the way a CD-ROM recorder works by letting you "burn" programs onto blanks once and then letting you read from them many times. In fact, programming a PROM is also called *burning*, just like burning a CD-R, and it is comparable in terms of its flexibility.
- **Erasable Programmable ROM (EPROM):** An *EPROM* is a ROM that can be erased and reprogrammed. A little glass window is installed in the top of the ROM package, through which you can actually see the chip that holds the memory. Ultraviolet light of a specific frequency can be shined through this window for a specified period of time, which will erase the EPROM and allow it to be reprogrammed again.
- **Electrically Erasable Programmable ROM (EEPROM):** The next level of erasability is the *EEPROM*, which can be erased under software control. This is the most flexible type of ROM, and is now commonly used for holding BIOS programs.

b) RANDOM-ACCESS MEMORY (RAM)

RAM is another type of memory found inside computers. It may be compared to a chalkboard on which we can scribble down notes, read them, and erase them when finished. In the computer, RAM is the working memory. Data can be read (retrieved) or written (stored) in RAM by providing the computer with an address location where the data is stored or where you want it to be stored. When the data is no longer required, we may simply write over it. Thus you can use the storage location again for something else.

The RAM family includes two important memory devices **static RAM** (SRAM) and **dynamic RAM** (DRAM). The primary difference between them is the lifetime of the data they store. *SRAM* retains its contents as long as electrical power is applied to the chip. If the power is turned off or lost temporarily, its contents will be lost forever. *DRAM*, on the other hand, has an extremely short data lifetime-typically about four milliseconds. This is true even when power is applied constantly.

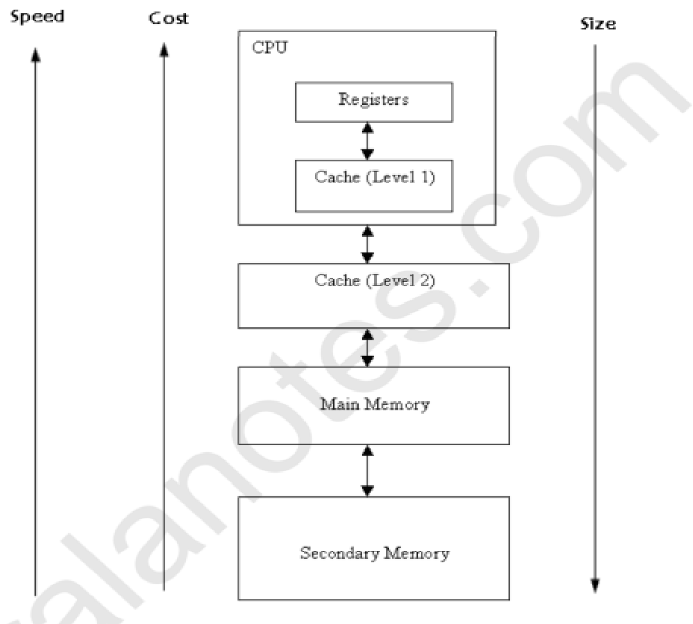
A simple piece of hardware called a *DRAM controller* can be used to make DRAM behave more like SRAM. The job of the DRAM controller is to periodically refresh the data stored in the DRAM. By refreshing the data before it expires, the contents of memory can be kept alive for as long as they are needed. So DRAM is as useful as SRAM after all.

SRAM devices offer extremely fast access times (approximately four times faster than DRAM) but are much more expensive to produce. Generally, SRAM is used only where access speed is extremely important. A lower cost-per-byte makes DRAM attractive whenever large amounts of RAM are required.

Secondary storage, or auxiliary storage, is memory external to the main body of the computer (CPU) where programs and data can be stored for future use. When the computer is ready to use these programs, the data is read into primary storage. Secondary storage media extends the storage capabilities of the computer system. Secondary storage is required for two reasons. First, the working memory of the CPU is limited in size and cannot always hold the amount of data required. Second, data and programs in secondary programs do not disappear when the power is turned off. Secondary storage is nonvolatile memory. This information is lost only when we erase it. Magnetic disks are the most common type of secondary storage. They may be either floppy disks or hard disks (hard drives).

20.

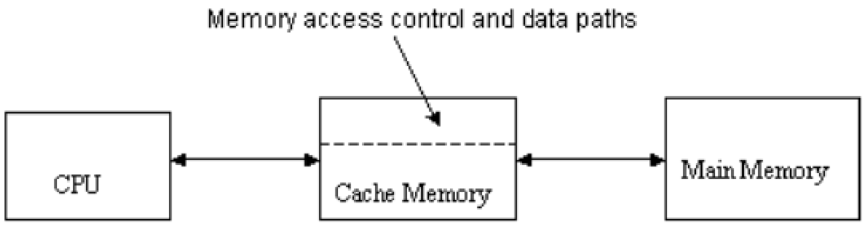
Draw the picture of memory hierarchy.



21

What is the need of a cache memory?

Speed of main memory is very low compared to processor speed. The speed gap between processor and memory is resolved by using cache memories. The cache memory is a small and fast memory that is placed between the main memory and the CPU. It holds the currently active and more frequently used segments of program and its data.



In this approach, the CPU need not always fetch the program and data from the main memory instead it can get it from the cache memory. Hence, the total time of execution is considerably reduced and it is more economical.

22.

Explain the basic principle of cache.

The cache working is based on the phenomenon **Locality of Reference.**

Principle of Locality of reference: Program access a relatively small portion of the address space at any instant of time.

Example: 90% of time in 10% of the code

The locality of reference occur in two ways

- **Spatial Locality**

Instructions are closed to recently executed instruction (based on their address) are likely to be executed soon.

- **Temporal Locality**

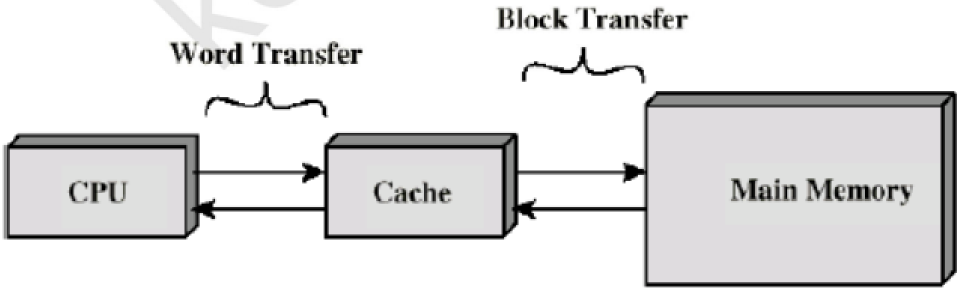
Recently executed instructions are likely to be executed again very soon.

If the active segments are placed in the cache then execution time can be reduced then the total execution time is reduced. Due to temporal aspect, wherever information (data) are first needed this item should be brought into cache. Due to spatial aspect, instead of fetching an item fetch the assignment also.

23.

. Explain the basic operation of cache.

The cache memory is divided into number of blocks or cache lines.



When the CPU requests for an addressed word, it is transferred from the main memory to the cache. The memory-access control circuitry in the cache, checks whether the requested word currently exists in the cache. If the requested word exists in cache, then the **Read or Write hit** occurs and the required operation is performed on the appropriate location in the cache. When the operation is Read, the main memory is not involved. However, if the operation is Write, then there are two techniques followed. In the first technique known as **write through**, the main memory and cache locations are updated simultaneously. In the second technique known as **write back**, only the cache location is updated and it is marked as updated using **dirty** or **modified bit.** The main memory will be updated during swapping. If the requested word does not exist in cache, then **a read or write miss** (cache miss) occurs. If the operation is Read**,** then there are two techniques followed. In the first technique the block of words that contain the requested word is loaded from the main memory into the cache and the requested word is forwarded to the CPU. In the second technique the requested word is directly sent to the CPU as soon as it is read from main memory. This technique is known as **load-through or early restart.** This reduces the processor waiting time. When the operation is Write, and the requested word is not in cache, then it is written directly into the main memory.

24.

Define Hit Ratio of cache memory.

$$\text{Hit Ratio} = \frac{\text{Number of hits}}{\text{Total CPU references}}$$

$$\text{Hit Ratio} = \frac{\text{Number of hits}}{\text{Number of Hits} + \text{Number of Misses}}$$

25.

What are the mapping functions commonly used in cache memory?

Direct mapping technique

Associative mapping technique

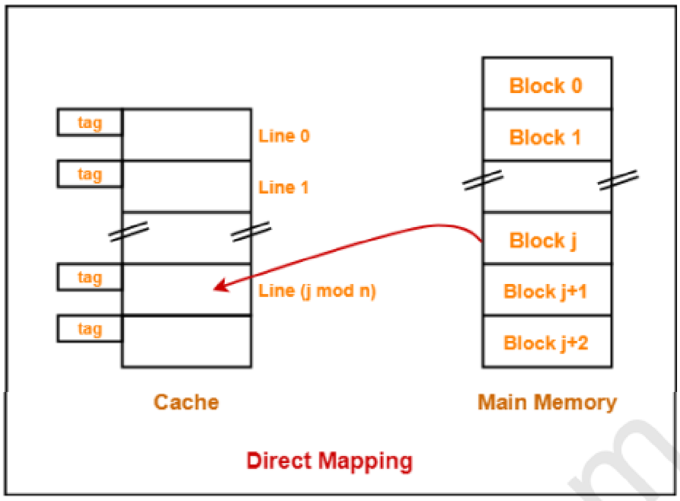
Set associative mapping technique

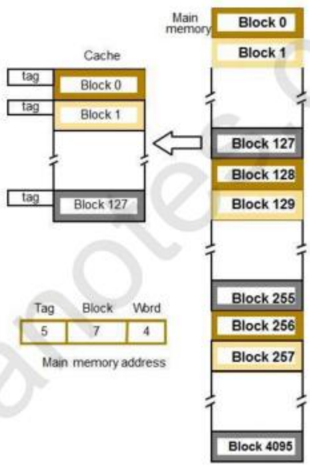
26.

Explain direct mapping technique.

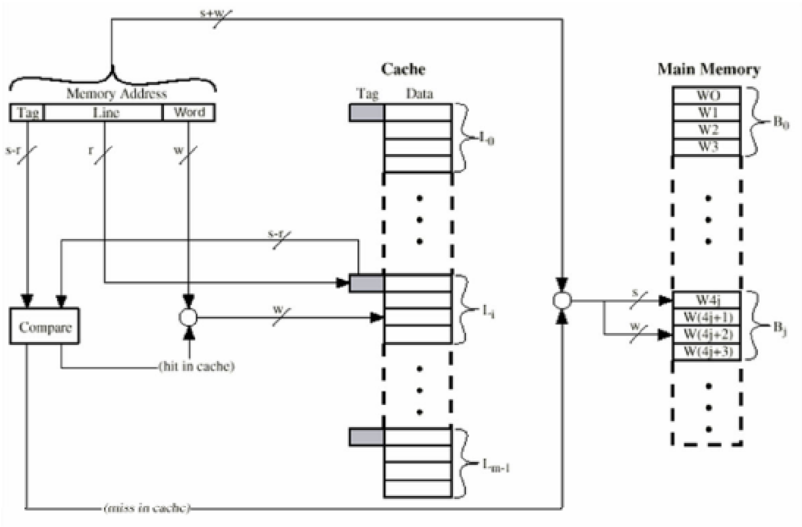
In direct mapped cache each main memory block is assigned to a specific line in the cache according to the formula i = j modulo N, where i is the cache line number assigned to main memory block j, and N is the number of lines in the cache. Direct mapping cache interprets a main memory address (comprising of s + w bits) as 3 distinct fields. Tag (the most significant s - r bits) is a unique identifier for each memory block. Line number (middle r bits) specifies which line will hold the referenced address. Line number identify one of the $N = 2^r$ lines of cache.

Word (least significant w bits) specifies the offset of a byte within a line or block. Word specifies the specific byte in a cache line that is to be accessed.





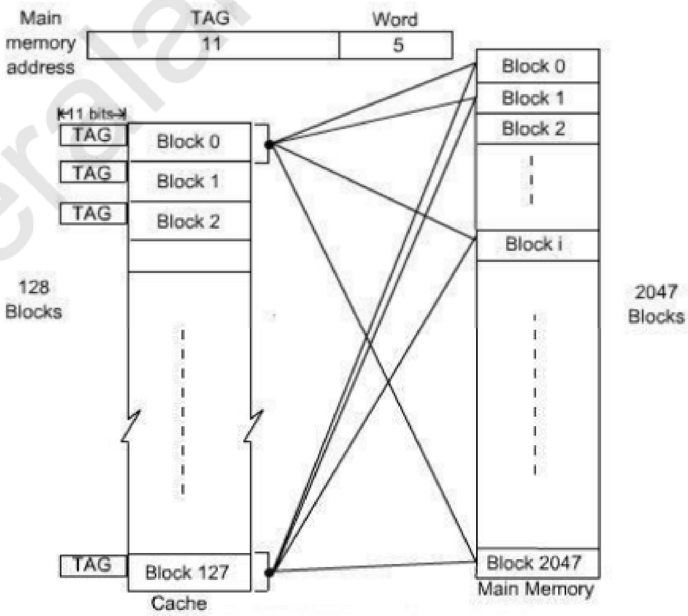
Block j of the main memory maps to j modulo 128 of the cache. (ie) Block 0, 128, 256 of main memory is maps to block 0 of cache memory. 1,129,257 maps to 1, & so on.



27.

Explain Associative mapping.

In associative mapping a block of the main memory can map to any available cache line. Associative mapping cache interprets a main memory address (comprising of s + w bits) as two distinct fields. Tag (the most significant s bits) is a unique identifier for each memory block. Word (least significant w bits) specifies the offset of a byte within a line or block. It might not be practical to use this complete flexibility of associative mapping technique due to searching overhead, because the TAG field of main memory address has to be compared with the TAG field of the entire cache block.

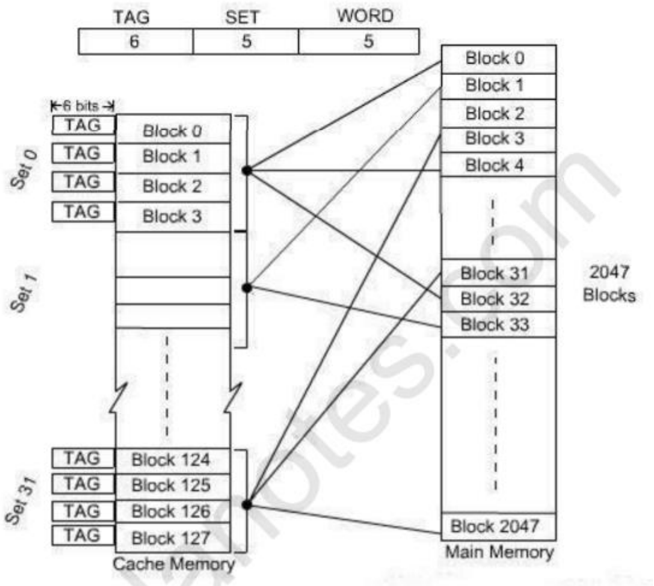


28.

. **Explain Set associative mapping.**

This mapping is compromise between direct mapping and fully associative mapping. It builds on the strengths of both: namely, the easy control of the direct mapped cache and

the more flexible mapping of the fully associative cache. In set associative mapping, cache is divided into a number of smaller direct-mapped areas called sets (v), with each set holding a number of lines (k).The cache is then described in the number of lines each set contains. If a set can hold X lines, the cache is referred to as an X-way set associative cache. A main memory block can be stored in any one of the k lines in a set such that *cache set number* = *j modulo v (*where *j* is the main memory block number).



The TAG field of associative mapping technique is divided into two groups, one is termed as SET bit and the second one is termed as TAG bit. Each set contains 4 blocks, total number of set is 32. The main memory address is grouped into three parts: low-order 5 bits are used to identifies a word within a block. Since there are total 32 sets present, next 5 bits are used to identify the set. High-order 6 bits are used as TAG bits.

29.

. **What is meant by block replacement?**

When the cache is full, there is a need for replacement algorithm for replacing the cache block with a new block.

30.

). Discuss various block replacement algorithms in cache.

Three types of replacement algorithms

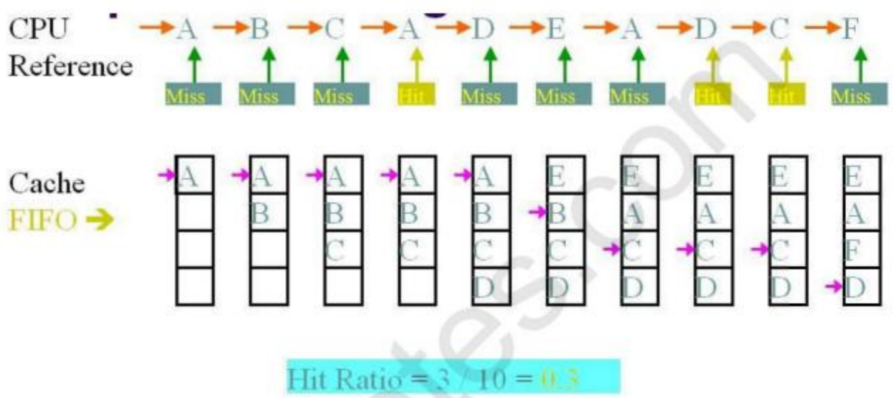
• Random replacement policy.

Choose the block to be overwritten at random. In this algorithm replace any cache line by using random selection. Simple and very effective in practice.

• First in first Out (FIFO) replacement policy

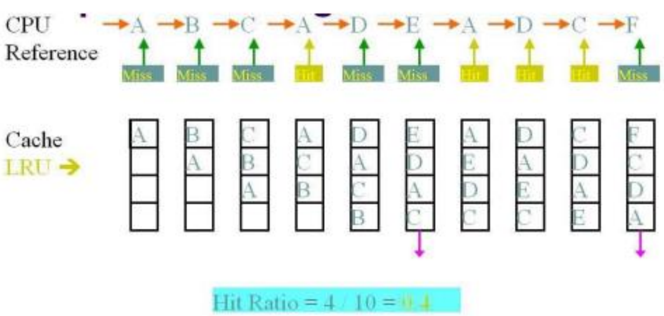
In this algorithm replace the cache block which is having the longest time stamp. While using this technique there is no need of updating when a hit occurs but when there is a miss occur then the block is put into an empty block and the counter values are incremented by one.

Example:



• Least recently used (LRU) replacement policy.

In the LRU, replace the cache block which is having the less reference with the longest time stamp. In this case also when a hit occurs when the counter value will be set to 0 but when the miss occurs there will be arising of two possibilities in which one possibility is that counter value is set as 0 and in another possibility, the counter value can be incremented as 1.

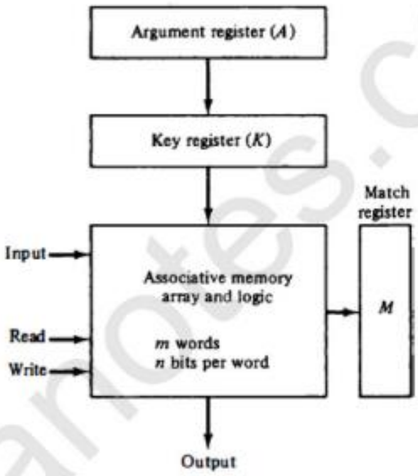


31.

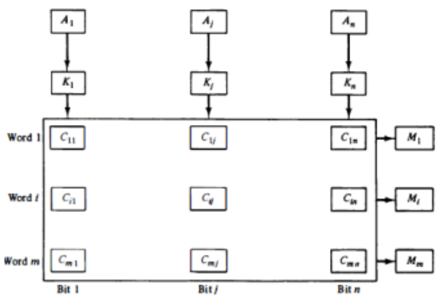
. Discuss the concept of associative memory.

A memory unit accessed by content is called **an associative memory or Content Addressable Memory (CAM).**

This type of memory is accessed simultaneously and in parallel on the basis of data content rather than by specific address or location. When a word is written in an associative memory, no address is given. The memory is capable of finding an empty unused location to store the word. When a word is to be read from an associative memory, the content of the word, or part of the word, is specified. The memory locates all words, which match the specified content and marks them for reading. The organization of associative memory is as shown in figure 8. The comperand register is also known as argument register. The masking register is known as key register.



The match register M has m bits, one for each memory word. Each word in memory is compared in parallel with the content of the argument register. The words that match the bits of the argument register set a corresponding bit in the match register. After the matching process, those bits in the match register that have been set indicate the fact that their corresponding words have been matched.



Example

A	101 111100	
K	111 000000	
Word 1	100 111100	no match
Word 2	101 000001	match

32.

Write notes on Vectored Interrupts.

In a computer, a vectored interrupt is an I/O interrupt that tells the part of the computer that handles I/O interrupts at the hardware level that a request for attention from an I/O device has been received and and also identifies the device that sent the request.

A vectored interrupt is an alternative to a polled interrupt , which requires that the interrupt handler poll or send a signal to each device in turn in order to find out which one sent the interrupt request.

33.

Differentiate between synchronous and asynchronous buses.

Synchronous bus:

- Transmitter and receivers are synchronized of clock.
- Data bits are transmitted with synchronization of clock.
- Character is received at constant Rate.
- Data transfer takes place in block.
- Start and stop bit are required to establish communication of each character.
- Used in high – speed transmission.

Asynchronous bus:

- Transmitters and receivers are not synchronized by clock.
- Bit's of data are transmitted at constant rate.
- Character may arrive at any rate at receiver.
- Data transfer is character oriented.
- Start and stop bits are required to establish communication of each character.
- Used in low – speed transmission.

34.

Briefly explain static memory.

SRAM (**static RAM**) is random access memory (**RAM**) that retains data bits in its memory as long as power is being supplied. Unlike dynamic **RAM** (DRAM), which stores bits in cells consisting of a capacitor and a transistor, **SRAM** does not have to be periodically refreshed.

Advantages:

- Low power consumption
- Simplicity – a refresh circuit is not needed
- Reliability

Disadvantages:

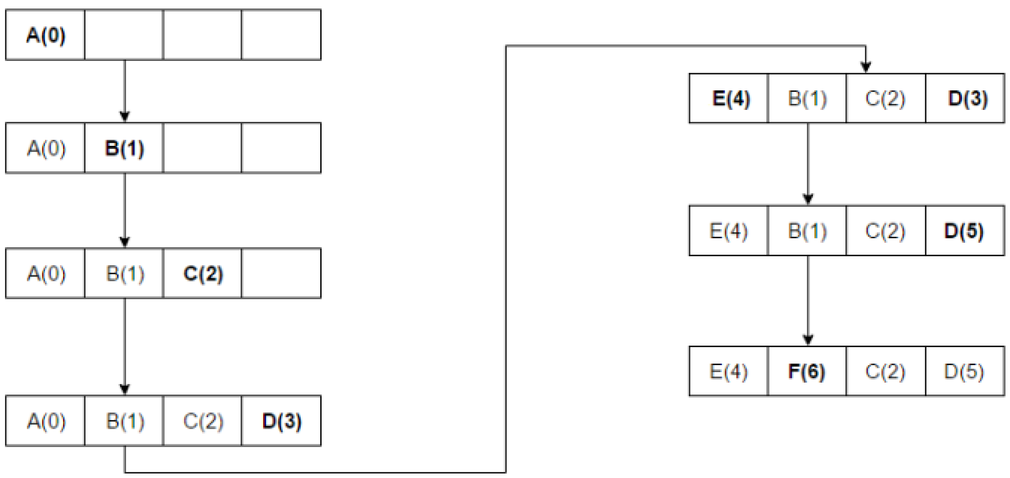
- Price
- Capacity

35.

Describe the LRU algorithm for cache replacement.

Discards the least recently used items first. This algorithm requires keeping track of what was used when, which is expensive if one wants to make sure the algorithm always discards *the* least recently used item. General implementations of this technique require keeping "age bits" for cache-lines and track the "Least Recently Used" cache-line based on age-bits. In such an implementation, every time a cache-line is used, the age of all other cache-lines changes.

The access sequence for the below example is A B C D E D F.



In the above example once A B C D gets installed in the blocks with sequence numbers (Increment 1 for each new Access) and when E is accessed, it is a miss and it needs to be installed in one of the blocks. According LRU Algorithm, since A has the lowest Rank(A(0)), E will replace A.

36.

Which are the different bus arbitration schemes?

There are two approaches to bus arbitration: Centralized and distributed.

1. Centralized Arbitration

- In centralized bus arbitration, a single bus arbiter performs the required arbitration. The bus arbiter may be the processor or a separate controller connected to the bus.
- There are three different arbitration schemes that use the centralized bus arbitration approach. There schemes are:

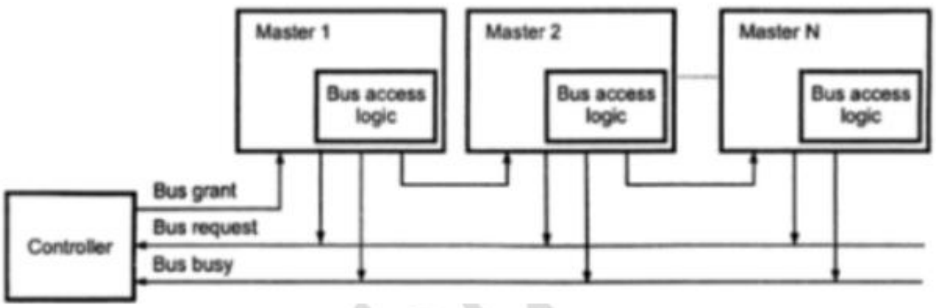
  a. Daisy chaining

  b. Polling method

  c. Independent request

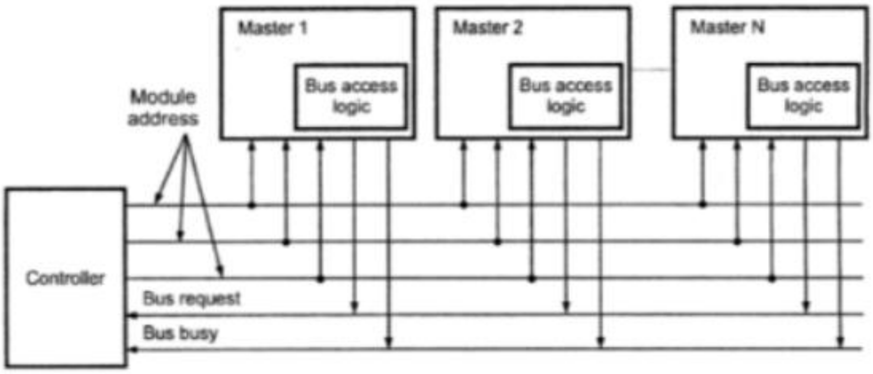
a) Daisy chaining

- The system connections for Daisy chaining method are shown in fig below.



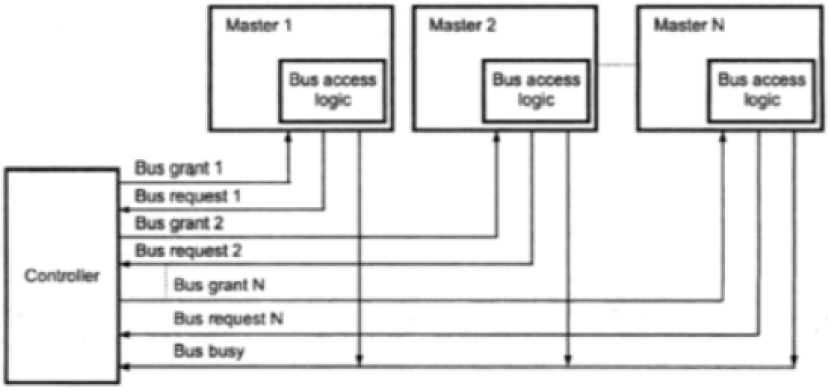
- It is simple and cheaper method. All masters make use of the same line for bus request.
- In response to the bus request the controller sends a bus grant if the bus is free.
- The bus grant signal serially propagates through each master until it encounters the first one that is requesting access to the bus. This master blocks the propagation of the bus grant signal, activities the busy line and gains control of the bus.
- Therefore any other requesting module will not receive the grant signal and hence cannot get the bus access.

b) Polling method



- The system connections for polling method are shown in figure above.
- In this the controller is used to generate the addresses for the master. Number of address line required depends on the number of master connected in the system.
- For example, if there are 8 masters connected in the system, at least three address lines are required.
- In response to the bus request controller generates a sequence of master address. When the requesting master recognizes its address, it activated the busy line ad begins to use the bus.

c) Independent request



- The figure below shows the system connections for the independent request scheme.
- In this scheme each master has a separate pair of bus request and bus grant lines and each pair has a priority assigned to it.
- The built in priority decoder within the controller selects the highest priority request and asserts the corresponding bus grant signal.

2. Distributed Arbitration

- In distributed arbitration, all devices participate in the selection of the next bus master.
- In this scheme each device on the bus is assigned a4-bit identification number.
- The number of devices connected on the bus when one or more devices request for the control of bus, they assert the start-arbitration signal and place their 4-bit ID numbers on arbitration lines, ARB0 through ARB3.

- These four arbitration lines are all open-collector. Therefore, more than one device can place their 4-bit ID number to indicate that they need to control of bus. If one device puts 1 on the bus line and another device puts 0 on the same bus line, the bus line status will be 0. Device reads the status of all lines through inverters buffers so device reads bus status 0as logic 1. Scheme the device having highest ID number has highest priority.
- When two or more devices place their ID number on bus lines then it is necessary to identify the highest ID number on bus lines then it is necessary to identify the highest ID number from the status of bus line. Consider that two devices A and B, having ID number 1 and 6, respectively are requesting the use of the bus.
- Device A puts the bit pattern 0001, and device B puts the bit pattern 0110. With this combination the status of bus-line will be 1000; however because of inverter buffers code seen by both devices is 0111.
- Each device compares the code formed on the arbitration line to its own ID, starting from the most significant bit. If it finds the difference at any bit position, it disables its drives at that bit position and for all lower-order bits.
- It does so by placing a 0 at the input of their drive. In our example, device detects a different on line ARB2 and hence it disables its drives on line ARB2, ARB1 and ARB0. This causes the code on the arbitration lines to change to 0110. This means that device B has won the race.
- The decentralized arbitration offers high reliability because operation of the bus is not dependent on any single device.

37.

Write notes on flash memory.

Flash Memory (sometimes called "**Flash RAM**") is a type of RAM that, like a ROM, retains its contents when the power supply is removed, but whose contents can be easily erased by applying a short pulse of higher voltage. This is *called flash erasure,* hence the name. Flash memory is currently both too expensive and too slow to serve as MAIN MEMORY, but is used as removable storage cards for digital cameras and pocket computers.

It is a variation of electrically erasable programmable read-only memory (EEPROM) which, unlike flash memory, is erased and rewritten at the byte level, which is slower than flash memory updating. Flash memory is often used to hold control code such as the basic input/output system (BIOS) in a personal computer. When BIOS needs to be changed (rewritten), the flash memory can be written to in block (rather than byte) sizes, making it easy to update. On the other hand, flash memory is not useful as random access memory (RAM) because RAM needs to be addressable at the byte (not the block) level.

Flash Memory gets its name because the microchip is so organized that a section of memory cells are erased in a single action or "flash." The erasure is caused by tunneling in which electrons pierce through a thin dielectric material to remove an electronic charge from a floating gate associated with each memory cell. Intel offers a form of flash memory that holds two bits (rather than one) in each memory cell, thus doubling the capacity of memory without a corresponding increase in price. Flash memory is used in digital cellular phones, digital cameras, LAN switches, PC Cards for notebook computers, digital set-up boxes, embedded controllers, and other devices.

USB, short for Universal Serial Bus, is a standard type of connection for many different kinds of devices.

Generally, USB refers to the types of cables and connectors used to connect these many types of external devices to computers.

The Universal Serial Bus standard has been extremely successful. USB ports and cables are used to connect hardware such as printers, scanners, keyboards, mice, flash drives, external hard drives, joysticks, cameras, and more to computers of all kinds, including desktops, tablets, laptops, netbooks, etc.

In fact, USB has become so common that you'll find the connection available on nearly any computer-like device such as video game consoles, home audio/visual equipment, and even in many automobiles.

Many portable devices, like smartphones, ebook readers, and small tablets, use USB primarily for charging. USB charging has become so common that it's now easy to find replacement electrical outlets at home improvement stores with USB ports built it, negating the need for a USB power adapter.

USB Versions

There have been three major USB standards, 3.1 being the newest:

- USB 3.1: Called *Superspeed+*, USB 3.1 compliant devices are able to transfer data at 10 Gbps (10,240 Mpbs).
- USB 3.0: Called *SuperSpeed USB*, USB 3.0 compliant hardware can reach a maximum transmission rate of 5 Gbps (5,120 Mbps).
- USB 2.0: Called *High-Speed USB*, USB 2.0 compliant devices can reach a maximum transmission rate of 480 Mbps.
- USB 1.1: Called *Full Speed USB*, USB 1.1 devices can reach a maximum transmission rate of 12 Mbps.

How USB Works

When a computer is powered up and USB devices are connected to a hub, the system will query and request from them the information on how much bandwidth is needed. Enumeration process will then occur where each device is assigned with a unique address. After that, the system will determine what kind of data the USB devices wish to transfer. There will be 4 different modes of transfers:

- Interrupt: used for devices which transfer little amount of data but need fast response (E.g. mouse, keyboard)
- Bulk: used for devices which receive big packet of data (E.g. printer)
- Isochronous: used for devices which requires streaming process (E.g. speaker, webcam)
- Control: short, simple commands to the device, and a status response.

After all connected devices are enumerated, the computer system will take care of the overall bandwidth and allocate it to different devices according to their transfer mode. Most of the bandwidth will be used for interrupt and isochronous transfer to ensure their requests are guaranteed. Once 90% of bandwidth is taken, the computer will refuse any other transfer from these two modes. Bulk or control

transfer (if available) will then take up the remaining bandwidth amount, which is up to 10%.

USB Main Features

- A maximum of 127 peripherals can be connected to a single USB host controller.
- USB device has a maximum speed up to 480 Mbps (for USB 2.0).
- Length of individual USB cable can reach up to 5 meters without a hub and 40 meters with hub.
- USB acts as "plug and play" device.
- USB can draw power by its own supply or from a computer. USB devices use power up to 5 voltages and deliver up to up to 500 mA.
- If a computer turns into power-saving mode, some USB devices will automatically convert themselves into "sleep" mode

38.

Describe the different types of DRAMs.

DRAM types

The different types of DRAM are used for different applications as a result of their slightly varying properties. The different types are summarised below:

- ***Asynchronous DRAM:*** Asynchronous DRAM is the basic type of DRAM on which all other types are based. Asynchronous DRAMs have connections for power, address inputs, and bidirectional data lines.

  Although this type of DRAM is asynchronous, the system is run by a memory controller which is clocked, and this limits the speed of the system to multiples of the clock rate. Nevertheless the operation of the DRAM itself is not synchronous.

  There are various types of asynchronous DRAM within the overall family:

  - *RAS only Refresh, ROR:* This is a classic asynchronous DRAM type and it is refreshed by opening each row in turn. The refresh cycles are spread across the overall refresh interval. An external counter is required to refresh the rows sequentially.
  - *CAS before RAS refresh, CBR:* To reduce the level of external circuitry the counter required for the refresh was incorporated into the main chip. This became the standard format for refresh of an asynchronous DRAM. (It is also the only form generally used with SDRAM).

- ***FPM DRAM:*** FPM DRAM or Fast Page Mode DRAM was designed to be faster than conventional types of DRAM. As such it was the main type of DRAM used in PCs, although it is now well out of date as it was only able to support memory bus speeds up to about 66 MHz.
- ***EDO DRAM:*** Extended Data Out DRAM was a form of DRAM that provided a performance increase over FPM DRAM. Yet this type of DRAM was still only able to operate at speeds of up to about 66 MHz.

  EDO DRAM is sometimes referred to as Hyper Page Mode enabled DRAM because it is a development of FPM type of DRAM to which it bears many similarities. The EDO DRAM type has the additional feature that a new access cycle could be started while the data output from the previous cycle was still present. This type of DRAM began its data output on the falling edge of /CAS line. However it did not inhibit the output when /CAS line rises. Instead, it held the output valid until either /RAS was dis-asserted, or a new /CAS falling edge selected a different column address. In some instances it was possible to carry out a memory transaction in one clock cycle, or provide an improvement from using three clock cycles to two dependent upon the scenario and memory used.

  This provided the opportunity to considerably increase the level of memory performance while also reducing costs.

- ***BEDO DRAM:*** The Burst EDO DRAM was a type of DRAM that gave improved performance of the straight EDO DRAM. The advantage of the BEDO DRAM type is that it could process four memory addresses in one burst saving three clock cycles when compared to EDO memory. This was done by adding an on-chip address counter count the next address.

BEDO DRAM also added a pipelined to enable the page-access cycle to be devided in to two components

1. the first component accessed the data from the memory array to the output stage
2. the second component drove the data bus from this latch at the appropriate logic level

Since the data was already in the output buffer, a faster access time is achieved - up to 50% improvement when compared to conventional EDO DRAM.

BEDO DRAM provided a significant improvement over previous types of DRAM, but by the time it was introduced, SDRAM had been launched and took the market. Therefore BEDO DRAM was little used.

- ***SDRAM:*** Synchronous DRAM is a type of DRAM that is much faster than previous, conventional forms of RAM and DRAM. It operates in a synchronous mode, synchronising with the bus within the CPU.
- ***RDRAM:*** This is Rambus DRAM - a type of DRAM that was developed by Rambus Inc, obviously taking its name from the company. It was a competitor to SDRAM and DDR SDRAM, and was able to operate at much faster speeds than previous versions of DRAM.

39.

Compare the speed, size and cost of different types of memories.

Dynamic RAM (DRAM)

Each memory cell in a DRAM chip holds one bit of data and is composed of a transistor and a capacitor. The transistor functions as a switch that allows the control circuitry on the memory chip to read the capacitor or change its state, while the capacitor is responsible for holding the bit of data in the form of a 1 or 0.

In terms of function, a capacitor is like a container that stores electrons. When this container is full, it designates a 1, while a container empty of electrons designates a 0. However, capacitors have a leakage that causes them to lose this charge, and as a result, the "container" becomes empty after just a few milliseconds.

Thus, in order for a DRAM chip to work, the CPU or memory controller must recharge the capacitors that are filled with electrons (and therefore indicate a 1) before they discharge in order to retain the data. To do this, the memory controller reads the data and then rewrites it. This is called refreshing and occurs thousands of times per second in a DRAM chip. This is also where the "Dynamic" in Dynamic RAM originates, since it refers to the refreshing necessary to retain the data.

Because of the need to constantly refresh data, which takes time, DRAM is slower.

Static RAM (SRAM)

Static RAM, on the other hand, uses flip-flops, which can be in one of two stable states that the support circuitry can read as either a 1 or a 0. A flip-flop, while requiring six transistors, has the advantage of not needing to be refreshed. The lack of a need to constantly refresh makes SRAM faster than DRAM; however, because SRAM needs more parts and wiring, an SRAM cell takes up more space on a chip than a DRAM cell does. Thus, SRAM is more expensive, not only because there is less memory per chip (less dense) but also because they are harder to manufacture.

Speed

Because SRAM does not need to refresh, it is typically faster. The average access time of DRAM is about 60 nanoseconds, while SRAM can give access times as low as 10 nanoseconds.

Capacity and Density

Because of its structure, SRAM needs more transistors than DRAM to store a certain amount of data. While a DRAM module only requires one transistor and one capacitor to store every bit of data, SRAM needs 6 transistors. Since the number of transistors in a memory module determines its capacity, for a similar number of transistors, a DRAM module can have up to 6 times more capacity than an SRAM module.

Power Consumption

Typically, an SRAM module consumes less power than a DRAM module. This is because SRAM only requires a small steady current while DRAM requires bursts of power every few milliseconds to refresh. This refresh current is several orders of magnitude greater than the low SRAM standby current. Thus, SRAM is used in most portable and battery-operated equipment.

However, the power consumption of SRAM does depend on the frequency at which it is accessed. When SRAM is used at a slower pace, it draws nearly negligible power while idled. On the other hand, at higher frequencies, SRAM can consume as much power as DRAM.

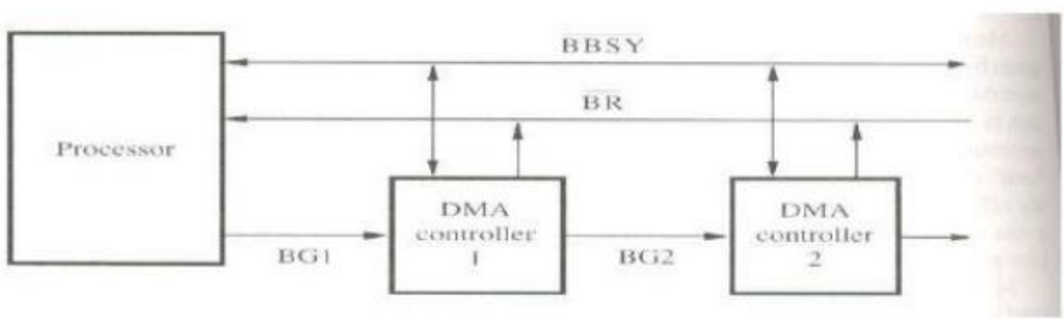
Price

SRAM is much more expensive than DRAM. A gigabyte of SRAM cache costs around $5000, while a gigabyte of DRAM costs $20-$75. Since SRAM uses flip-flops, which can be made of up to 6 transistors, SRAM needs more transistors to store 1 bit than DRAM does, which only uses a single transistor and capacitor. Thus, for the same amount of memory, SRAM requires a higher number of transistors, which increases the production cost.

40.

Draw the simple arrangement of bus arbitration using daisy chaining in DMA.

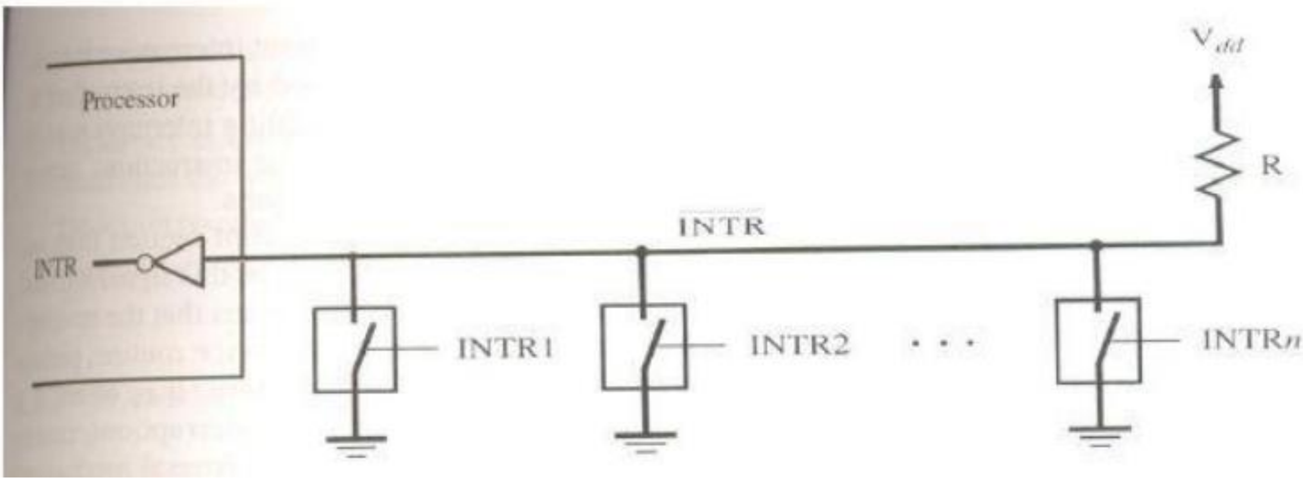
Fig:A simple arrangement for bus arbitration using a daisy chain



- Here the processor is the bus master and it may grants bus mastership to one of its DMA controller.
- A DMA controller indicates that it needs to become the bus master by activating the Bus Request line (BR) which is an open drain line.
- The signal on BR is the logical OR of the bus request from all devices connected to it.
- When BR is activated the processor activates the Bus Grant Signal (BGI) and indicated the DMA controller that they may use the bus when it becomes free.
- This signal is connected to all devices using a daisy chain arrangement.
- If DMA requests the bus, it blocks the propagation of Grant Signal to other devices and it indicates to all devices that it is using the bus by activating open collector line, Bus Busy (BBSY).

41.

Write notes on hardware interrupts with help of diagram.



- A single interrupt request line may be used to serve „n‟ devices. All devices are connected to the line via switches to ground.
- To request an interrupt, a device closes its associated switch, the voltage on INTR line drops to 0(zero).
- If all the interrupt request signals (INTR1 to INTRn) are inactive, all switches are open and the voltage on INTR line is equal to Vdd.
- When a device requests an interrupts, the value of INTR is the logical OR of the requests from individual devices.
- (ie) **INTR = INTR1+………..+INTRn**
- **INTR** □ It is used to name the INTR signal on common line it is active in the low voltage state.
- **Open collector** (bipolar ckt) or **Open drain** (MOS circuits) is used to drive INTR line.
- The Output of the Open collector (or) Open drain control is equal to a switch to the ground that is open when gates input is in „0‟ state and closed when the gates input is in „1‟ state.
- Resistor „R‟ is called a **pull-up resistor** because it pulls the line voltage upto the high voltage state when the switches are open.

42.

Write pros and cons of SRAM.

Merit of SRAM

- It has low power consumption because the current flows in the cell only when the cell is being activated accessed.
- Static RAM‟s can be accessed quickly. It access time is few nanoseconds.

Demerit of SRAM

- SRAM‟s are said to be volatile memories because their contents are lost when the power is interrupted

43.

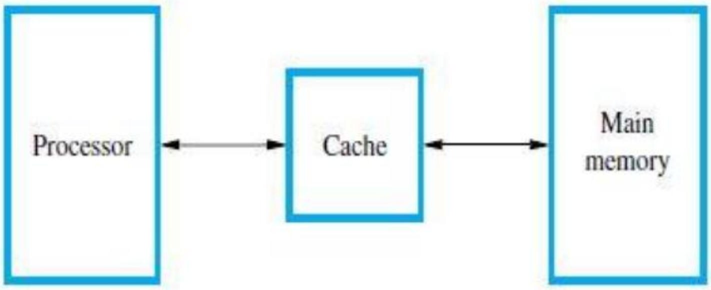
Explain DDRSRAM.

- The standard SDRAM performs all actions on the rising edge of the clock signal.
- The double data rate SDRAM transfer data on both the edges(loading edge, trailing edge).
- The Bandwidth of DDR-SDRAM is doubled for long burst transfer.
- To make it possible to access the data at high rate , the cell array is organized into two banks.
- Each bank can be accessed separately.
- Consecutive words of a given block are stored in different banks.
- Such interleaving of words allows simultaneous access to two words that are transferred on successive edge of the clock.

44.

Explain cache memory.

- The effectiveness of cache mechanism is based on the property of computer program called '**Locality of reference'.**
- **Locality of Reference:**
- Many instructions in the localized areas of the program are executed repeatedly during some time period and remainder of the program is accessed relatively infrequently.
- It manifests itself in 2 ways. They are,
- Ø **Temporal**(The recently executed instruction are likely to be executed again very soon.)
- Ø **Spatial**(The instructions in close proximity to recently executed instruction are also likely to be executed soon.)
- The term **Block** refers to the set of contiguous address locations of some size.
- The **cache line**is used to refer to the cache block



- The Cache memory stores a reasonable number of blocks at a given time but this number is small compared to the total number of blocks available in Main Memory.

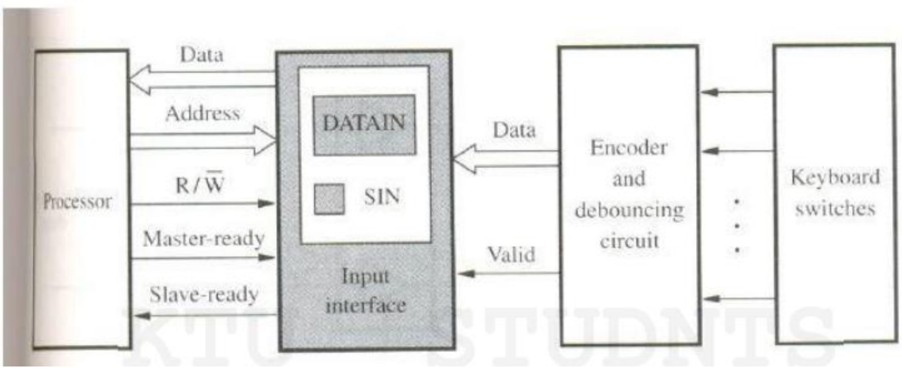
- The correspondence between main memory block and the block in cache memory is specified by a mapping function.
- The Cache control hardware decide that which block should be removed to create space for the new block that contains the referenced word.
- The collection of rule for making this decision is called the **replacement algorithm.**
- The cache control circuit determines whether the requested word currently exists in the cache.
- If it exists, then Read/Write operation will take place on appropriate cache location. In this case **Read/Write hit** will occur.
- In a Read operation, the memory will not involve

- The write operation is proceed in 2 ways. They are,
- Ø Write-through protocol
- Ø Write-back protocol
- **Write-through protocol:**
- Here the cache location and the main memory locations are updated simultaneously.
- **Write-back protocol:**
- This technique is to update only the cache location and to mark it as with associated flag bit called **dirty/modified bit.**
- The word in the main memory will be updated later, when the block containing this marked word is to be removed from the cache to make room for a new block.
- If the requested word currently not exists in the cache during read operation, then **read miss** will occur.
- To overcome the read miss **Load –through / Early restart protocol** is used

45.

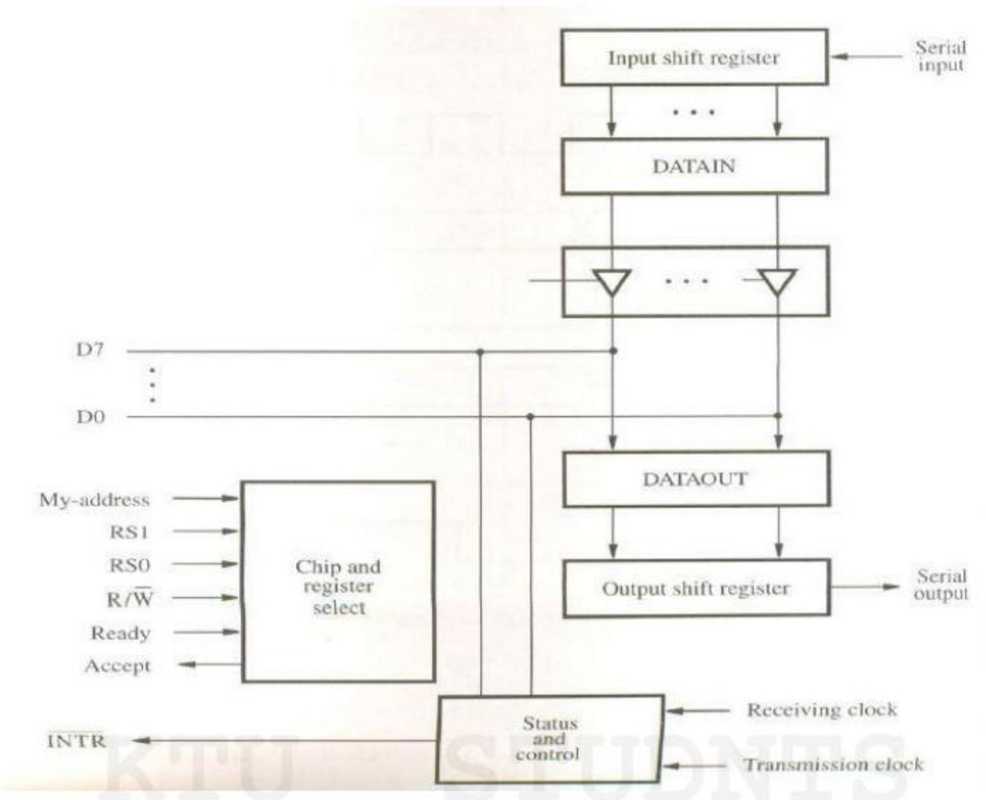
What are the two types of interface circuits? Explain with the help of a figure.

- Parallel Port
- Serial Port



- The output of the encoder consists of the bits that represent the encoded character and one signal called **valid**,which indicates the key is pressed.
- The information is sent to the interface circuits,which contains a data register,DATAIN and a status flag SIN.
- When a key is pressed, the Valid signal changes from 0 to1,causing the ASCII code to be loaded into DATAIN and SIN set to 1.
- The status flag SIN set to 0 when the processor reads the contents of the DATAIN register.
- The interface circuit is connected to the asynchronous bus on which transfers are controlled using the Handshake signals Master ready and Slave-ready.

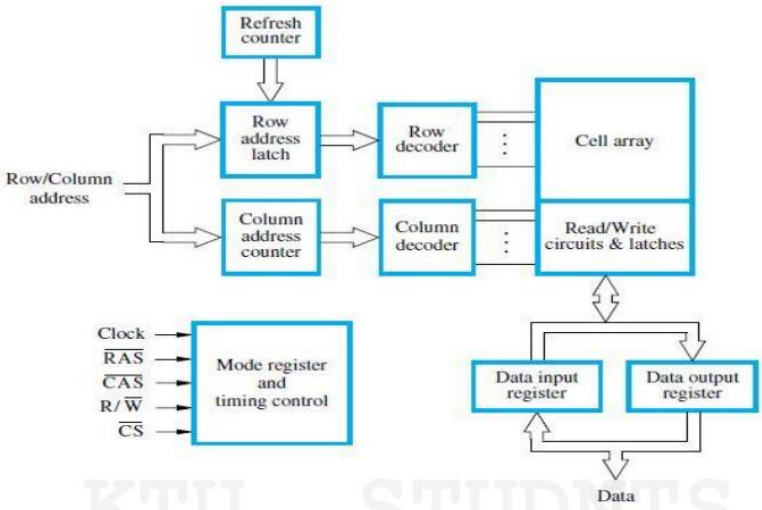
Serial Port



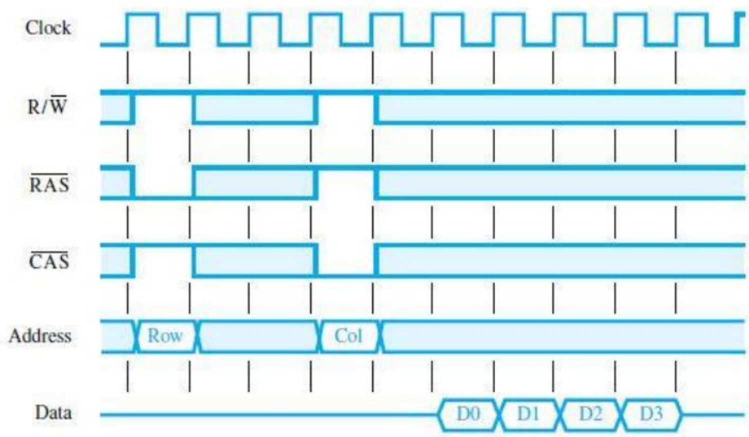
- A serial port used to connect the processor to I/O device that requires transmission one bit at a time.
- It is capable of communicating in a bit serial fashion on the device side and in a bit parallel fashion on the bus side.

46.

Explain synchronous DRAM with the help of a figure.



- Here the operations directly synchronized with clock signal.
- The address and data connections are buffered by means of registers.
- The output of each sense amplifier is connected to a latch.
- A Read operation causes the contents of all cells in the selected row to be loaded in these latches.
- Data held in the latches that correspond to the selected columns are transferred into the data output register, thus becoming available on the data output pins.



- First ,the row address is latched under control of RAS signal.
- The memory typically takes 2 or 3 clock cycles to activate the selected row.
- Then the column address is latched under the control of CAS signal.
- After a delay of one clock cycle, the first set of data bits is placed on the data lines.
- The SDRAM automatically increments the column address to access the next 3 sets of bits in the selected row, which are placed on the data lines in the next 3 clock cycles.

47. List and explain the different types of ROMs .

48. Explain the term locality of reference. How is this related to cache memory?

49. Explain with examples the three types of mapping functions used in cache memory.

50. What are interrupts? Outline the actions taking place in a processor once an interrupt has been raised.

51. What is a DRAM? Compare the two types of DRAMs, highlighting their differences.

52. Outline how Direct Memory Access is implemented? Differentiate between cycle stealing DMA and burst mode DMA.